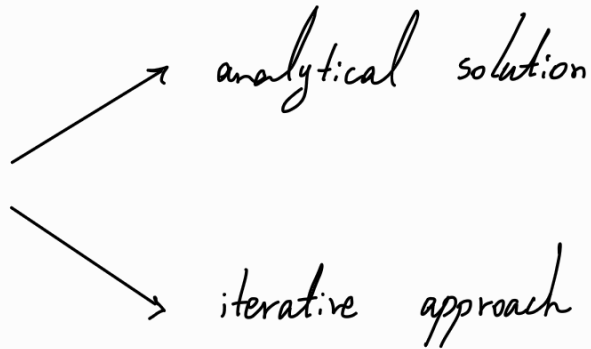


Policy Evaluation

Suppose π is a deterministic stationary policy that we aim to evaluate, i.e., compute its value function.

$$\rightarrow \vec{v}^\pi(s) \quad \forall s \in \mathcal{S}$$



Analytical solution: According to the Bellman consistency equation

$$\begin{aligned} v^\pi(s) &= \mathbb{E}_{a \sim \pi(s)} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [v^\pi(s')] \right] \\ &= R(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} [v^\pi(s')] \\ &= R(s, \pi(s)) + \gamma \sum_{s' \in \mathcal{S}} P(s' | s, \pi(s)) v^\pi(s') \quad \forall s \in \mathcal{S} \end{aligned}$$

For a finite discrete state space $\mathcal{S} = \{1, 2, \dots, n\}$, we can rewrite

this in vector form :

$$\vec{v}^\pi = \begin{bmatrix} v^\pi(1) \\ v^\pi(2) \\ \vdots \\ v^\pi(n) \end{bmatrix}_{|\mathcal{S}| \times 1} \quad \vec{R}^\pi = \begin{bmatrix} R(1, \pi(1)) \\ R(2, \pi(2)) \\ \vdots \\ R(n, \pi(n)) \end{bmatrix}_{|\mathcal{S}| \times 1} \quad P^\pi = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}_{|\mathcal{S}| \times |\mathcal{S}|}$$

$$\begin{aligned}
P_{i,j}^{\pi} &= \mathbb{P}[s_{t+1}=j \mid s_t=i] \\
&= \sum_{a \in \mathcal{A}} \mathbb{P}[s_{t+1}=j, A_t=a \mid s_t=i] \\
&= \sum_{a \in \mathcal{A}} \mathbb{P}[s_{t+1}=j \mid s_t=i, A_t=a] \mathbb{P}[A_t=a \mid s_t=i] \\
&= \sum_{a \in \mathcal{A}} P(j|i,a) \pi(a|i) \\
&\xrightarrow[\text{policy}]{\text{deterministic}} P_{i,j}^{\pi} = P(j|i, \pi(i))
\end{aligned}$$

Therefore, we have

$$\vec{v}^{\pi}(s) = \vec{R}^{\pi}(s) + \gamma \langle \overbrace{P^{\pi}(s)}^{\text{s}^{\text{th}} \text{ row of } P^{\pi}}, \vec{v}^{\pi} \rangle$$

$$\begin{bmatrix} \square \\ \vdots \end{bmatrix} = \begin{bmatrix} \square \\ \vdots \end{bmatrix} + \gamma \begin{bmatrix} \text{---} \\ \vdots \end{bmatrix} \begin{bmatrix} | \\ \vdots \end{bmatrix}$$

$$\Rightarrow \vec{v}^{\pi} = \vec{R}^{\pi} + \gamma P^{\pi} \vec{v}^{\pi}$$

$$\longrightarrow (I - \gamma P^{\pi}) \vec{v}^{\pi} = \vec{R}^{\pi} \longrightarrow \vec{v}^{\pi} = (I - \gamma P^{\pi})^{-1} \vec{R}^{\pi}$$

$\nearrow$ row stochastic
 $\nwarrow$ $0 < \gamma < 1$

Note: The computational complexity of the analytical solution is $O(|\mathcal{S}|^3)$ when using basic matrix inversion techniques.

Iterative approach: This approach relies on the fixed-point iteration algorithm.

Fixed-point solution: For a function $f: D_f \rightarrow \mathbb{R}$, $x \in D_f$ is a fixed point if $f(x) = x$. $\rightarrow f(f(x)) = x \rightarrow (f \circ f \circ \dots \circ f)(x) = x$

Fixed-point iteration algorithm: An iterative approach to find a fixed-point solution.

- start with $x_0 \in D_f$
- Update with $x_{t+1} = f(x_t) \quad \forall t \in \mathbb{N}_0$

Banach Fixed-point Theorem: A contraction mapping over a non-empty and complete metric space has a unique fixed-point solution.

The sequence of points from the fixed-point iteration algorithm converges to this solution.

Contraction mapping: A function $f(\cdot)$ is a contraction mapping if it satisfies the contraction property, i.e.,

$$\forall x_1, x_2 \in D_f : \|f(x_1) - f(x_2)\| \leq \alpha \|x_1 - x_2\|$$

for some $0 \leq \alpha < 1$.

f Lipschitz
with constant < 1

Approximate Policy Evaluation

- Initialize $\vec{v}_0$
- For $t = 0, 1, 2, \dots, T-1$:

$$\vec{v}_{t+1} = \vec{R}^\pi + \gamma P^\pi \vec{v}_t$$

- Return $\vec{v}_T$
- $\curvearrowright \approx \vec{v}^\pi$

fixed-point iteration algorithm for

$$f(\vec{v}) = \vec{R}^\pi + \gamma P^\pi \vec{v}$$

Lemma: let $\vec{v}_0, \vec{v}_1, \dots, \vec{v}_T$ represent the value function at different iterations of the algorithm. It holds that:

$$\| \underbrace{\vec{v}_{t+1}}_{f(\vec{v}_t)} - \underbrace{\vec{v}^\pi}_{f(\vec{v}_2)} \|_\infty \leq \gamma \| \underbrace{\vec{v}_t}_{\vec{v}_1} - \underbrace{\vec{v}^\pi}_{\vec{v}_2} \|_\infty$$

Proof:

$$\begin{bmatrix} \text{---} \\ P^\pi \end{bmatrix} \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

$\langle \dots \rangle$

$$= \| \gamma P^\pi (\vec{v}_t - \vec{v}^\pi) \|_\infty = \gamma \| P^\pi (\vec{v}_t - \vec{v}^\pi) \|_\infty$$

$$= \gamma \max_{s \in \mathcal{S}} | [P^\pi (\vec{v}_t - \vec{v}^\pi)]_s |$$

$$= \gamma \max_{s \in \mathcal{S}} | \langle P^\pi(s), \vec{v}_t - \vec{v}^\pi \rangle |$$

g is convex



↓

$g(\mathbb{E}[x]) \leq \mathbb{E}[g(x)]$ Jensen's inequality

$$= \gamma \max_{s \in \mathcal{S}} \left| \mathbb{E}_{s' \sim p_{\pi}(s)} [\vec{V}_t - \vec{V}^{\pi}]_{s'} \right|$$

$$\leftarrow \leq \gamma \max_{s \in \mathcal{S}} \mathbb{E}_{s' \sim p_{\pi}(s)} [|\vec{V}_t(s') - \vec{V}^{\pi}(s')|]$$

avg $\leq$ max $\leftarrow$

$$\leq \gamma \max_{s' \in \mathcal{S}} |\vec{V}_t(s') - \vec{V}^{\pi}(s')|$$

$$= \gamma \|\vec{V}_t - \vec{V}^{\pi}\|_{\infty}$$

■

Theorem: For any accuracy level $\epsilon > 0$, if we run this iterative algorithm for $T \geq \frac{\log(\frac{\|\vec{V}_0 - \vec{V}^{\pi}\|_{\infty}}{\epsilon})}{\log \frac{1}{\gamma}}$, then, $\|\vec{V}_T - \vec{V}^{\pi}\|_{\infty} \leq \epsilon$.

Proof: Based on the previous lemma, after T iterations

$$\begin{aligned} \|\vec{V}_T - \vec{V}^{\pi}\|_{\infty} &\leq \gamma \|\vec{V}_{T-1} - \vec{V}^{\pi}\|_{\infty} \\ &\leq \gamma^2 \|\vec{V}_{T-2} - \vec{V}^{\pi}\|_{\infty} \\ &\vdots \\ &\leq \gamma^T \|\vec{V}_0 - \vec{V}^{\pi}\|_{\infty} \end{aligned}$$

We want $\|\vec{V}_T - \vec{V}^*\|_\infty \leq \epsilon$. It will hold if

$$\gamma^T \|\vec{V}_0 - \vec{V}^*\|_\infty \leq \epsilon \implies \gamma^T \leq \frac{\epsilon}{\|\vec{V}_0 - \vec{V}^*\|_\infty}$$

$$\implies T \log \gamma \leq \log \frac{\epsilon}{\|\vec{V}_0 - \vec{V}^*\|_\infty}$$

$$\implies T \geq \frac{\log \left(\frac{\|\vec{V}_0 - \vec{V}^*\|_\infty}{\epsilon} \right)}{\log \frac{1}{\gamma}}$$

Note: The computational complexity of the iterative approach is $O(T|S|^2)$ which will be $O(|S|^2 \log(1/\epsilon))$ if ϵ -accuracy is desired.

If the reward is bounded, then the value function will be bounded.

Let $|R(s, a)| \leq R_{\max} \quad \forall s \in S, \forall a \in A \implies$

$$\begin{aligned} V^*(s) &= \mathbb{E}_{\pi, P} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, A_t) \mid s_0 = s \right] \\ &\leq \mathbb{E}_{\pi, P} \left[\sum_{t=0}^{\infty} \gamma^t R_{\max} \mid s_0 = s \right] = R_{\max} \sum_{t=0}^{\infty} \gamma^t = \frac{R_{\max}}{1-\gamma} \end{aligned} \quad \forall s \in S$$